

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

100 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

100 KWP GRID-TIED SOLAR PLANT

LT CATEGORY II — COMMERCIAL

Net investment ₹43.81 L after input tax credit, generating 1,53,300 units in year one against an energy charge of ₹10.15 per unit.

30-YEAR ANALYSIS

PAYBACK

3.2 yrs

Post-tax, undiscounted

PROJECT IRR

31.3%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.52

per kWh, against ₹10.15 from the grid

NET PRESENT VALUE

₹1.00 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	100 kWp
Year 1 generation	1,53,300 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	LT Category II — Commercial (LT supply)
Energy charge offset	₹10.15 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹14.43 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹46.00 / Wp (incl. GST)
Project price incl. GST	₹46.00 L
GST at 5% — recovered as ITC	₹2.19 L
Net capitalised cost	₹43.81 L
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹11.03 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	43,80,952	17,52,381	26,28,571	4,41,039
2	26,28,572	10,51,429	15,77,143	2,64,624
3	15,77,143	6,30,857	9,46,286	1,58,774
4	9,46,285	3,78,514	5,67,771	95,264
5	5,67,772	2,27,109	3,40,663	57,159
6	3,40,663	1,36,265	2,04,398	34,295
7	2,04,398	81,759	1,22,639	20,577
8	1,22,638	49,055	73,583	12,346
9	73,583	29,433	44,150	7,408
10	44,150	17,660	26,490	4,445
11	26,490	10,596	15,894	2,667
12	15,894	6,358	9,536	1,600
Tax shield, years 1-12				11,00,198

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30-year post-tax cash flow

100 kWp
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How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	1,53,300	14,42,553	—	17,52,381	(77,977)	15,20,530	15,20,530
2	1,52,534	14,92,754	60,000	10,51,429	95,972	13,36,782	28,57,312
3	1,51,771	15,44,702	60,000	6,30,857	2,14,896	12,69,806	41,27,118
4	1,51,012	15,98,457	60,000	3,78,514	2,91,934	12,46,523	53,73,641
5	1,50,257	16,54,084	60,000	2,27,109	3,44,040	12,50,043	66,23,684
6	1,49,506	17,11,646	60,000	1,36,265	3,81,391	12,70,255	78,93,939
7	1,48,758	17,71,211	60,000	81,759	4,10,100	13,01,111	91,95,050
8	1,48,014	18,32,849	60,000	49,055	4,33,844	13,39,005	1,05,34,055
9	1,47,274	18,96,632	60,000	29,433	4,54,836	13,81,796	1,19,15,851
10	1,46,538	19,62,635	60,000	17,660	4,74,411	14,28,225	1,33,44,076
11	1,45,805	20,30,935	60,000	10,596	4,93,378	14,77,557	1,48,21,633
12	1,45,076	21,01,611	60,000	6,358	5,12,233	15,29,379	1,63,51,012
13	1,44,351	21,74,747	60,000	3,815	5,31,280	15,83,468	1,79,34,480
14	1,43,629	22,50,429	60,000	2,289	5,50,711	16,39,718	1,95,74,198
15	1,42,911	23,28,744	60,000	1,373	5,70,652	16,98,092	2,12,72,290
16	1,42,196	24,09,784	60,000	824	5,91,186	17,58,598	2,30,30,888
17	1,41,485	24,93,644	60,000	494	6,12,375	18,21,269	2,48,52,157
18	1,40,778	25,80,423	60,000	297	6,34,265	18,86,158	2,67,38,315
19	1,40,074	26,70,222	60,000	178	6,56,896	19,53,326	2,86,91,641
20	1,39,374	27,63,146	60,000	107	6,80,301	20,22,845	3,07,14,486
21	1,38,677	28,59,303	60,000	64	7,04,512	20,94,791	3,28,09,277
22	1,37,983	29,58,807	60,000	38	7,29,562	21,69,245	3,49,78,522
23	1,37,293	30,61,773	60,000	23	7,55,480	22,46,293	3,72,24,815
24	1,36,607	31,68,323	60,000	14	7,82,299	23,26,024	3,95,50,839
25	1,35,924	32,78,581	60,000	8	8,10,050	24,08,530	4,19,59,369
26	1,35,244	33,92,675	60,000	5	8,38,766	24,93,909	4,44,53,278
27	1,34,568	35,10,740	60,000	3	8,68,482	25,82,259	4,70,35,537
28	1,33,895	36,32,914	60,000	2	8,99,231	26,73,683	4,97,09,220
29	1,33,226	37,59,339	60,000	1	9,31,049	27,68,290	5,24,77,510
30	1,32,560	38,90,164	60,000	1	9,63,976	28,66,189	5,53,43,699
Total	42,80,620	7,42,23,827	17,40,000	43,80,952	1,71,40,131	5,53,43,699	5,53,43,699

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

100 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.2 yrs
Discounted payback	4.1 yrs
Project IRR, unlevered	31.3%
Net present value at 10%	₹1.00 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.52 / kWh
Cumulative return, 30 years	1163%
Total net cash flow, after tax	₹5.53 Cr
30-year generation	42,80,621 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.52
Grid energy charge today		₹10.15

Every unit the plant generates costs **₹3.52** across its life against **₹10.15** bought from APSPDCL today — and the grid figure reaches **₹31.65** by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 42,80,621 units, avoiding 3,510.1 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** LT Category II — Commercial at ₹10.15/kWh. Marginal rate, above 500 units per month. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹46.00 L.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 100 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.